Course: Big Data & Management Tools

Date: 14-01-2026

Inventory of employer and union relationships in the Collective Agreements e-Library Portal

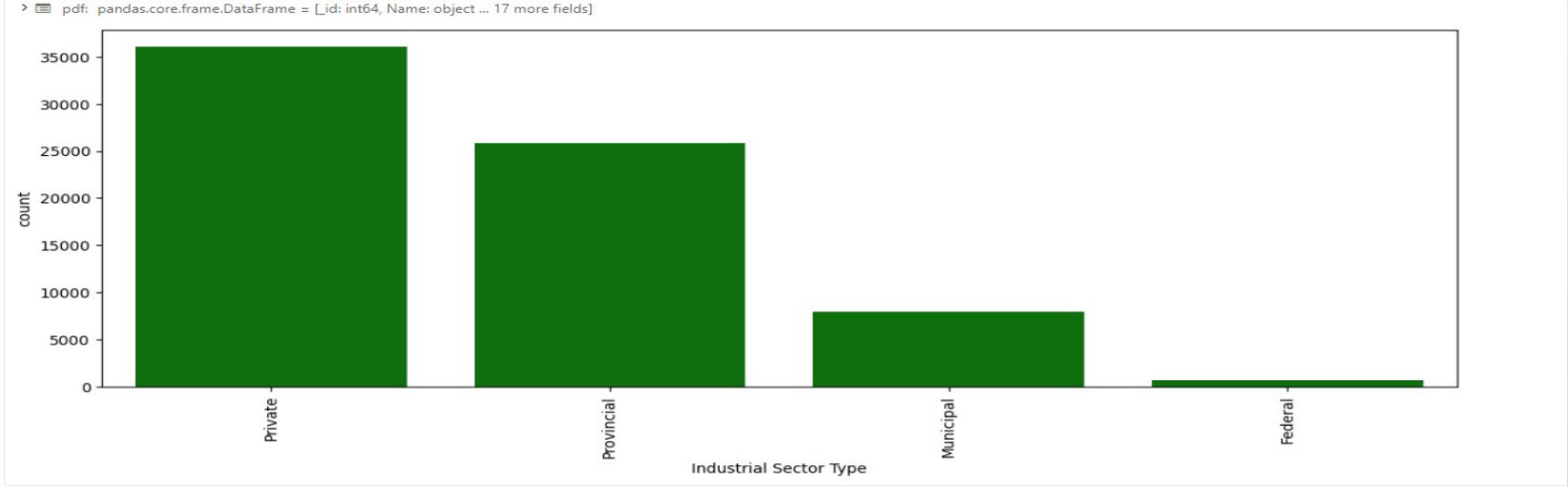
Introduction and Project Goal

- The Collective Agreements e-Library Portal contains detailed information on employer–union relationships across multiple industrial sectors.
- It contains 70,000+ records and 15 columns including ‘Employer’, ‘Union’, ‘Sector’.
- The dataset has 20777 unique employers, 9443 unique unions, 4 different industrial sector types.
- This project analyzes the dataset to identify underlying patterns in collective bargaining agreements using distributed data processing and machine learning techniques.

	_id	Name	Employer	Union	Agreement Start Date	Agreement Expiry Date	Industrial Sector Type	Labour Legislation (Act)	Number of Employees	Legacy Relationship ID	Industry	Sector	Agreement Location	OLRB Construction Industry Areas
0	1	019-0001-13.pdf	R.J.'s Tree Farms	International Brotherhood Of Teamsters, Local 879	2010-05-01	2013-04-30	Private	Labour Relations Act, 1995	0 - 50	30620	Agriculture and Natural Resources	Crop Production	Ontario - Province wide	NaN

Distribution of Sector Type

- The majority of collective agreements are concentrated in the private sector.
- Provincial agreements are significantly fewer than private, but still substantial, reflects centralized bargaining in education, healthcare, and public services.
- Municipal agreements are relatively limited, suggesting localized and standardized bargaining structures.
- Federal-level collective agreements are rare compared to other sectors.

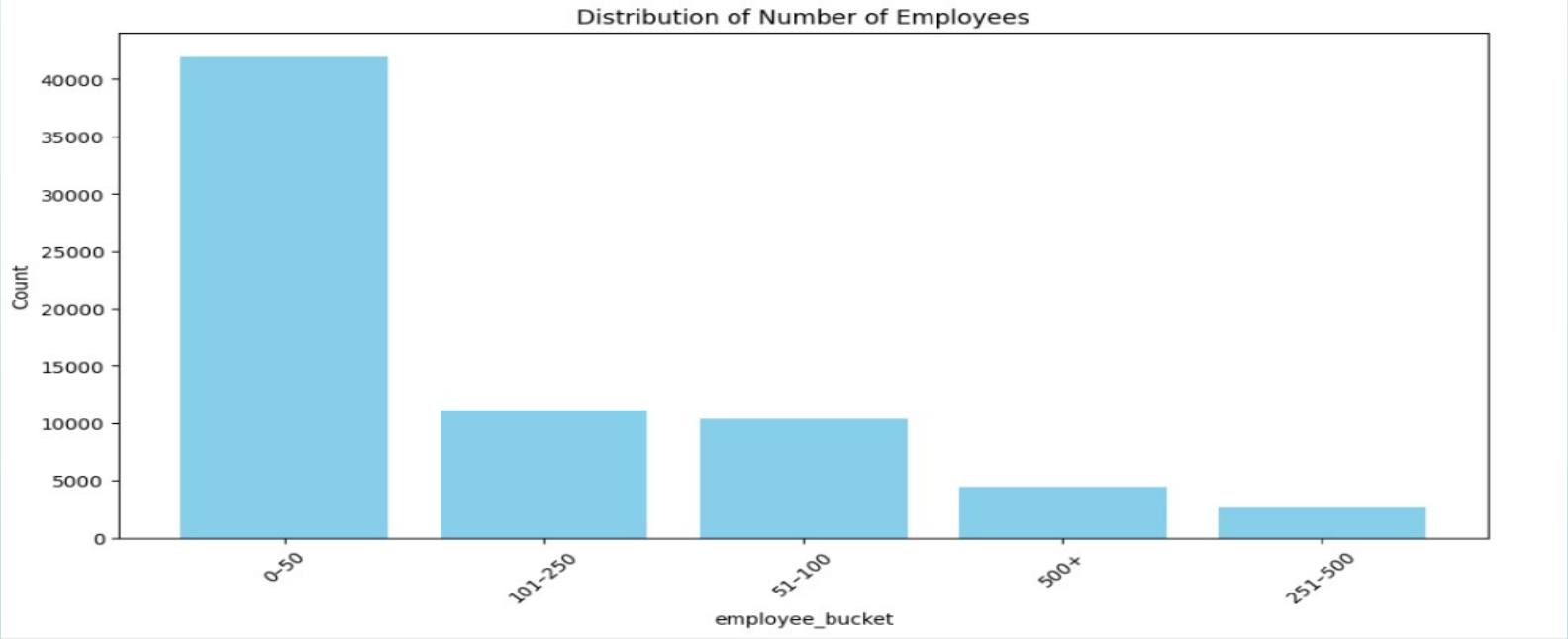


Contract Duration Patterns

- Around 50,000 Agreements have 25-50 months of contracts.
- And 15000+ agreements are 0-25 months of contracts in length.
- Very few agreements have 6+ years(75+ months) contract

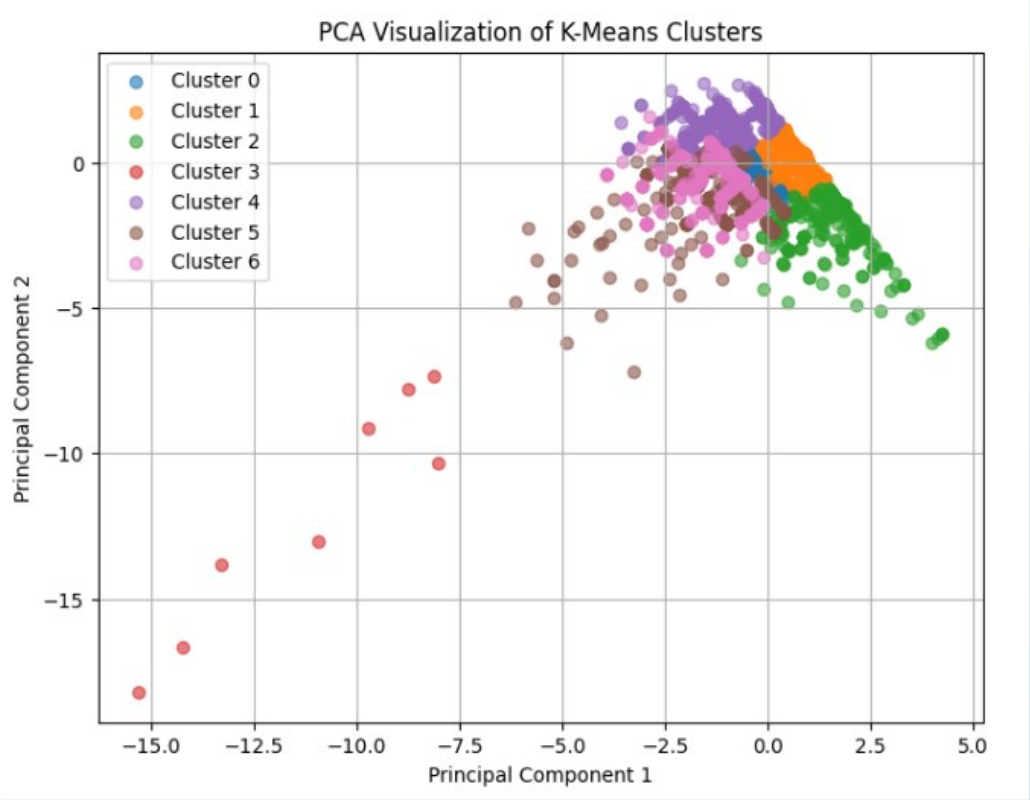
Distribution of Number of employees

- Most agreements involve small organizations (0–50 employees)
- Second most(10,000+) agreements involve medium-sized organization(101-250employees)



K-Means Cluster Model Development

- The optimal number of clusters was determined using the Silhouette score.
- PCA with two components was applied after clustering solely for visualization purpose.
- Some clusters were dominated by private-sector agreements with shorter contract durations.
- Cluster far left & bottom (large negative PC1 & PC2)
- This suggests that outlier group
- Possibly agreements with:
  - Very high or very low employee counts
  - Unusual contract lengths



Findings and Summary

- The majority of collective agreements belong to the private sector, followed by provincial, municipal, and federal sectors.
- Most agreements involve small to medium-sized employers, while large employers are less frequent.
- Contract durations vary widely, with certain sectors showing longer-term agreements.
- K-Means clustering identified distinct groups of collective agreements based on: Number of employees, Contract length, Industrial sector type.
- Other clusters represented public-sector agreements with larger workforces and longer contracts
- The final model demonstrated reasonable cluster separation, indicating meaningful grouping of agreement characteristics.
- Feature scaling significantly improved clustering stability and interpretability.

Limitations

- High-cardinality attributes such as agreement location were excluded from modeling.
- Name attribute contains file name/number and it’s excluded from analysis and modeling.

Thank You